

Standard Form & Indices

Test 7 · Answer sheet · 41 marks total

Section A — Laws of indices

A1. 9^9 (1)

A2. 4^5 (1)

A3. 8^{10} (1)

A4. $10^2 \times 10^1 = 10^3$ (1) = **1000** (1)

Section B — Zero and negative indices

B1. **1** (1)

B2. $1 \div 9^2 = 1 \div 81$ (1) = **1/81** (1)

B3. $1 \div 2^2 = 1 \div 4$ (1) = **0.25** (1)

Section C — Writing standard form

C1. 2.9 (1) = **2.9×10^4** (1)

C2. 4.5 (1) = **4.5×10^6** (1)

C3. 8.1 (1) = **8.1×10^{-3}** (1)

C4. 6 (1) = **6×10^{-7}** (1)

Section D — Converting back, and calculating

D1. **1,800,000** (1)

D2. **0.005** (1)

D3. $9 \times 4 = 36$, $10^2 \times 10^6 = 10^8$ (1) = 36×10^8 (1) adjust to standard form: **3.6×10^9** (1)

D4. $4 \div 8 = 0.5$, $10^8 \div 10^2 = 10^6$ (1) = 0.5×10^6 (1) adjust to standard form: **5×10^5** (1)

Section E — Powers to know, and estimating roots

E1. **625** (1)

E2. $6^2=36$, $7^2=49$ = between **6 and 7** (1)

Section F — Laws of indices with algebra

F1. **d^{15}** (1)

F2. **e^6** (1)

F3. **t^{16}** (1)

F4. $9 \times 2 = 18$ (1) $k^{4+3} = k^7$ (1) = **$18k^7$**

F5. $2^4 = 16$ (1) $w^{5 \times 4} = w^{20}$ (1) = **$16w^{20}$**

Section G — Fractional indices (Higher)

G1. $\sqrt{144} = \mathbf{12}$ (1)

G2. cube root of 343 = **7** (1)

G3. $\sqrt{9} = 3$ (1) $3^3 = \mathbf{27}$ (1)

G4. cube root of 1000 = 10, $10^2 = 100$ (1) $1 \div 100 = \mathbf{1/100}$ (1)

Marking note: award method marks for a correctly identified front number/power, or correctly applied law of indices, even if the final answer contains an arithmetic error. Accept equivalent fractions/decimals.